Why the Bounded-u_{ref} Run Produces a Very Small x_s

April 6, 2026

1 Question

The observed behavior is:

- in the *unbounded* frozen-$\hat{d}$ run, the online target state x_s is large,
- in the *bounded with* u_{ref} run, the online target state x_s stays close to zero,
- this raises the question of whether the steady-state equations were implemented incorrectly, or whether x_s is simply very dependent on u_s.

This note answers that question using the actual matrices and the latest two runs:

- bounded with u_{ref}: `Data/debug_exports/mpc_first_step_replacement_bounded_frozen_dhat/20260406_2`
- unbounded: `Data/debug_exports/mpc_first_step_replacement_unbounded_frozen_dhat/20260406_23295`

2 Bottom Line

The implementation is internally consistent. The evidence strongly supports:

1. the code is using the correct steady-state algebra,
2. x_s is exactly determined from u_s through

$$x_s = (I - A)^{-1} B u_s, \tag{1}$$

3. the very small bounded target state does *not* come from a wrong x_s formula,
4. it comes from the fact that the bounded-u_{ref} optimization is choosing a very small u_s.

The key structural point is:

$$\text{small bounded } x_s \quad \Longleftarrow \quad \text{small bounded } u_s \quad \Longleftarrow \quad \lambda_u \|u_s - u_{\text{ref}}\|_2^2 \text{ dominates } \|G u_s - r_k\|_2^2. \tag{2}$$

3 Steady-State Equations Used in the Code

For the frozen-$\hat{d}$ target, the code solves

$$(I - A)x_s - Bu_s = 0, \tag{3}$$
$$Cx_s = r_k, \tag{4}$$

where

$$r_k = y_{sp,k} - \hat{d}_k. \tag{5}$$

If $(I - A)$ is invertible, then

$$x_s = (I - A)^{-1}Bu_s, \tag{6}$$
$$Gu_s = r_k, \qquad G := C(I - A)^{-1}B. \tag{7}$$

Define

$$W := (I - A)^{-1}B. \tag{8}$$

Then

$$x_s = Wu_s, \qquad y_s - d_s = Cx_s = CWu_s = Gu_s. \tag{9}$$

This means:

- the output-side steady-state relation is controlled by G,
- the state-side steady-state relation is controlled by W.

4 Actual Identified Matrices

For the current notebook setup, the identified unaugmented model is

$$A = \begin{bmatrix} 0.946394 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0.936731 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0.885846 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.927830 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0.010342 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \tag{10}$$

$$B = \begin{bmatrix} 7.601178 \times 10^{-3} & 0 \\ 1.512532 \times 10^{-2} & 0 \\ 0 & 1.471516 \times 10^{-2} \\ 0 & 3.010827 \times 10^{-2} \\ 0 & 0 \\ 0 & 0 \\ 3.267612 \times 10^{-5} & 0 \end{bmatrix}, \tag{11}$$

$$C = \begin{bmatrix} 0 & 0 & 0.029531 & 0 & 1 & 0 & 0 \\ 0 & -0.050171 & 0 & 0.046562 & 0 & 0 & 0 \end{bmatrix}. \tag{12}$$

The matrix $(I - A)$ is not near singular:

$$\kappa(I - A) \approx 33.62. \tag{13}$$

So there is no evidence that the x_s calculation is blowing up because $(I - A)^{-1}$ is numerically invalid.

5 State Map and Output Map

Using the actual model,

$$W = (I - A)^{-1}B = \begin{bmatrix} 0.141797 & 0 \\ 0.239063 & 0 \\ 0 & 0.128906 \\ 0 & 0.417188 \\ 0.001499 & 0 \\ 0.001499 & 0 \\ 0.001499 & 0 \end{bmatrix}, \tag{14}$$

and

$$G = CW = \begin{bmatrix} 0.001499 & 0.003807 \\ -0.011994 & 0.019425 \end{bmatrix}. \tag{15}$$

Its inverse is

$$G^{-1} = \begin{bmatrix} 259.770673 & -50.907293 \\ 160.396561 & 20.047199 \end{bmatrix}. \tag{16}$$

5.1 Interpretation

These numbers immediately explain the behavior:

- W has moderate gain. The singular values of W are

  $$\sigma(W) = [0.436649,\ 0.277964]. \tag{17}$$

  So x_s is *not* exploding relative to u_s.

- The real amplification is in G^{-1}. A modest output-side right-hand side r_k can require a very large exact u_s.

So the main story is not

> "x_s is hypersensitive to u_s."

It is instead

> "the exact output-tracking target requires a huge u_s, and then x_s follows that u_s linearly through W."

6 Run-to-Run Comparison

Comparing the latest two runs gives:

Metric	bounded with u_{ref}	unbounded
mean $\|u_s\|_2$	5.0394	449.3150
mean $\|x_s\|_2$	1.7323	149.1337
ratio to unbounded	0.0112	1.0000
ratio of $\|x_s\|_2$	0.0116	1.0000

The crucial observation is that the shrinkage ratios are almost the same:

$$\frac{\mathbb{E}\|x_s^{\text{bounded}}\|}{\mathbb{E}\|x_s^{\text{unbounded}}\|} \approx 0.0116, \tag{18}$$

$$\frac{\mathbb{E}\|u_s^{\text{bounded}}\|}{\mathbb{E}\|u_s^{\text{unbounded}}\|} \approx 0.0112. \tag{19}$$

That is exactly what one expects from

$$x_s = W u_s \tag{20}$$

with a moderate-gain matrix W.

7 Direct Consistency Check

For both runs, the saved online target satisfies the steady-state state equation numerically:

$$x_s \approx W u_s. \tag{21}$$

The measured reconstruction residuals were:

$$\max_k \|x_s^{\text{bounded}}(k) - W u_s^{\text{bounded}}(k)\|_2 = 5.44 \times 10^{-16}, \tag{22}$$

$$\max_k \|x_s^{\text{unbounded}}(k) - W u_s^{\text{unbounded}}(k)\|_2 = 8.16 \times 10^{-14}. \tag{23}$$

Also, the run-to-run difference satisfies

$$x_s^{\text{unbounded}} - x_s^{\text{bounded}} = W\left(u_s^{\text{unbounded}} - u_s^{\text{bounded}}\right) \tag{24}$$

to machine precision:

$$\max_k \left\| \left(x_s^{\text{unbounded}} - x_s^{\text{bounded}}\right) - W\left(u_s^{\text{unbounded}} - u_s^{\text{bounded}}\right) \right\|_2 = 1.15 \times 10^{-13}. \tag{25}$$

This is very strong evidence that the implementation is algebraically correct.

8 Why the Bounded-u_{ref} Target Is So Small

The bounded-with-anchor objective is

$$\min_{u_s} \quad \|Gu_s - r_k\|_2^2 + \lambda_u \|u_s - u_{\text{ref}}\|_2^2 \qquad \text{subject to } u_{\min} \le u_s \le u_{\max}. \tag{26}$$

In the current run,

$$\lambda_u = 1.0, \qquad u_{\text{ref}} = u_{k-1}. \tag{27}$$

The Hessian of the output-tracking term is

$$G^\top G = \begin{bmatrix} 1.4610 \times 10^{-4} & -2.2728 \times 10^{-4} \\ -2.2728 \times 10^{-4} & 3.9182 \times 10^{-4} \end{bmatrix}, \tag{28}$$

with eigenvalues

$$\lambda(G^\top G) = [1.0604 \times 10^{-5},\ 5.2732 \times 10^{-4}]. \tag{29}$$

This is the crucial scale comparison:

$$G^\top G \sim 10^{-5} \text{ to } 10^{-4}, \qquad \lambda_u I \sim 1. \tag{30}$$

So the anchor term is about 10^3 to 10^5 times stronger than the output-tracking curvature.

8.1 Inactive Bounds in the Actual Run

In the bounded-with-u_{ref} run, the target never actually sat on the input bounds:

$$\text{fraction active at lower bounds} = [0,\ 0], \tag{31}$$

$$\text{fraction active at upper bounds} = [0,\ 0]. \tag{32}$$

So in this particular run, the small bounded target is *not* mainly caused by clipping on the box. It is mainly caused by the u_{ref} penalty.

8.2 Step-0 Proof

At step $k = 0$, the exact unbounded target was

$$u_s^{\text{unbounded}} = \begin{bmatrix} 689.123462 \\ 451.554842 \end{bmatrix}, \tag{33}$$

which corresponds to

$$r_0 = G u_s^{\text{unbounded}} = \begin{bmatrix} 2.752065 \\ 0.506106 \end{bmatrix}. \tag{34}$$

Because the bounded solution at this step is interior, the regularized unconstrained minimizer is

$$u_s^{\text{reg}} = (G^\top G + \lambda_u I)^{-1} (G^\top r_0 + \lambda_u u_{\text{ref}}). \tag{35}$$

With $u_{\text{ref}} = 0$ and $\lambda_u = 1$, this gives

$$u_s^{\text{reg}} = \begin{bmatrix} -0.001940 \\ 0.020300 \end{bmatrix}, \tag{36}$$

which matches the actual bounded online target:

$$u_s^{\text{bounded}} = \begin{bmatrix} -0.001939 \\ 0.020298 \end{bmatrix}. \tag{37}$$

This is the cleanest evidence in the whole analysis:

> the tiny bounded target is exactly what the weighted optimization asks for, even before the box constraints matter.

9 Why x_s Then Becomes Tiny

Once the optimization chooses a tiny u_s, the state target is forced to be tiny because

$$x_s = W u_s. \tag{38}$$

At step $k = 0$,

$$x_s^{\text{bounded}} = \begin{bmatrix} -2.75 \times 10^{-4} \\ -4.64 \times 10^{-4} \\ 2.62 \times 10^{-3} \\ 8.47 \times 10^{-3} \\ -2.91 \times 10^{-6} \\ -2.91 \times 10^{-6} \\ -2.91 \times 10^{-6} \end{bmatrix}, \tag{39}$$

$$x_s^{\text{unbounded}} = \begin{bmatrix} 97.715553 \\ 164.743578 \\ 58.208241 \\ 188.383036 \\ 1.033049 \\ 1.033049 \\ 1.033049 \end{bmatrix}. \tag{40}$$

This is not because x_s was coded incorrectly. It is because the chosen u_s changed by orders of magnitude, and x_s follows it linearly.

10 Answer

The correct answer is:

1. Yes, x_s is directly determined by u_s through the exact relation

$$x_s = (I - A)^{-1} B u_s. \tag{41}$$

2. No, the evidence does *not* suggest the code was written incorrectly.

3. The actual source of the discrepancy is the optimization scaling:
   - G^{-1} is large, so the exact output-tracking target wants a huge u_s,
   - W is moderate, so x_s then becomes proportionally large,
   - with $\lambda_u = 1$, the u_{ref} penalty dominates $G^\top G$,
   - therefore the bounded-with-u_{ref} optimization picks a tiny u_s,
   - and then $x_s = W u_s$ is also tiny.

11 Practical Consequence

If the goal is for the bounded target to remain closer to the exact frozen-$\hat{d}$ equilibrium, then the main tuning issue is not the x_s formula. The main tuning issue is the relative scale of

$$\|Gu - r_k\|_2^2 \quad \text{versus} \quad \lambda_u \|u - u_{\text{ref}}\|_2^2. \tag{42}$$

With the present model,

$$\lambda_u = 1 \tag{43}$$

is extremely large relative to the natural scale of $G^\top G$. That is why the bounded-with-anchor target stays very close to u_{ref} and therefore produces a very small x_s.